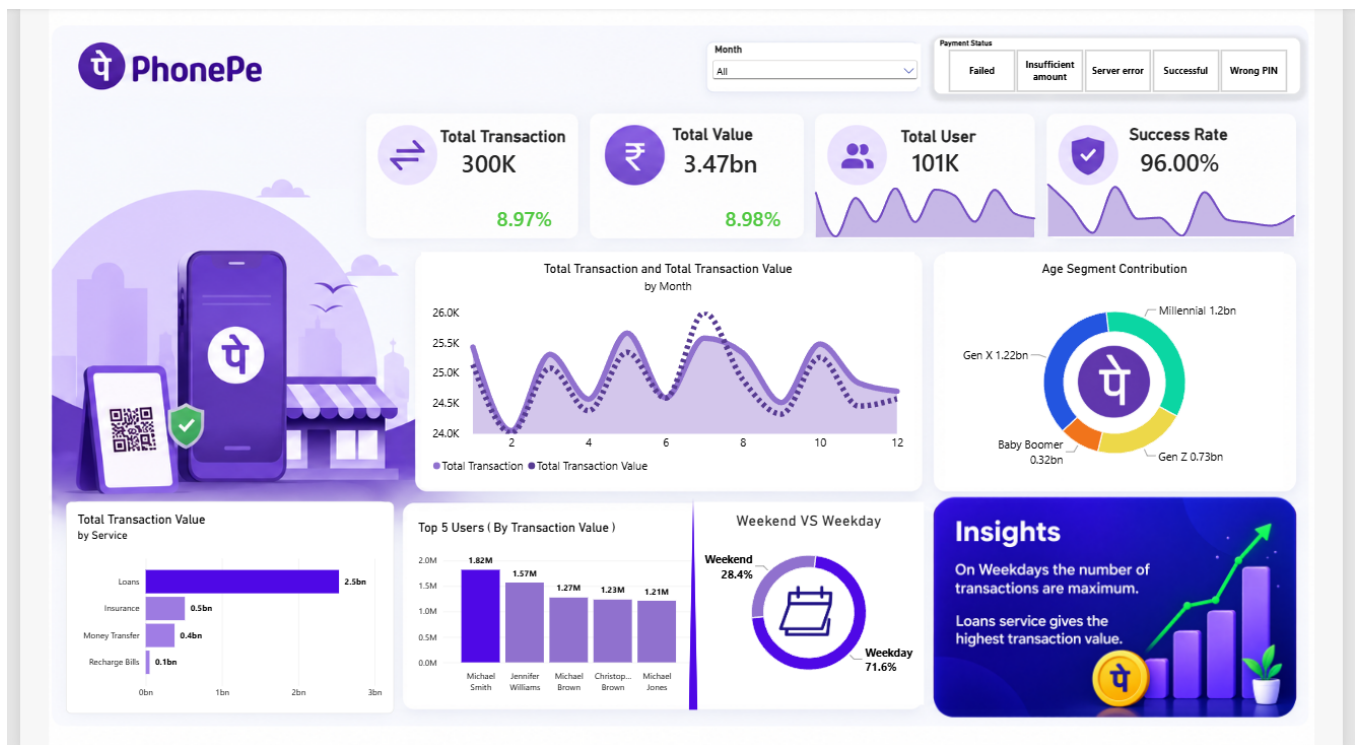


PHONEPE TRANSACTION ANALYTICS

Business problem, analytical approach and decision-ready outcomes



300K

Transactions

INR 3.47bn

Transaction value

100,761

Transacting users

96.0%

Success rate

Prepared by Tarun Yadav | Data Analytics Portfolio Project

Tools demonstrated: data cleaning, data modelling, DAX/KPI design, segmentation, trend analysis and business storytelling.

The business problem

A digital payments business handles large transaction volumes across multiple services, user groups and payment outcomes. Raw transaction records alone do not show where value is concentrated, when demand peaks, which customer segments matter most, or why payments fail.

Core question

How can PhonePe convert transaction and customer data into a single management view that improves service prioritisation, customer engagement and payment reliability?

Business objectives

- Create a trusted view of transaction volume, value, active users and payment success.
- Identify high-value services, peak periods and the customer cohorts that contribute most.
- Measure unsuccessful payments and reveal operational causes that require attention.
- Give decision-makers interactive filters for month and payment status.
- Translate findings into practical actions for operations, product and marketing teams.

Scope and source data

Users table	107,658 records User ID, name, age and join date
Transactions table	300,000 records Transaction ID, amount, user ID, service, service type, payment status, reason and date
Analysis period	1 January 2024 to 30 December 2024
Relationship	Users joined to transactions using User ID

How the analysis solved it

1. Profile

Checked row counts, missing values, duplicate IDs, dates and numeric validity.

2. Prepare

Standardised service and status fields; derived month, weekday/weekend and age segment.

3. Model

Built a user-to-transaction relationship on User ID for demographic analysis.

4. Measure

Calculated transaction count, transaction value, unique users and success rate.

5. Analyse

Compared services, months, day type, customer segments and payment outcomes.

6. Visualise

Designed one interactive Power BI view with KPI cards, slicers and focused charts.

7. Recommend

Converted patterns into actions for product reliability, capacity and targeting.

Data quality checks completed

Check	Result
Missing values	0 across both tables
Duplicate transaction IDs	0
Duplicate user IDs	0
Zero or negative amounts	0
Unmatched transaction users	0

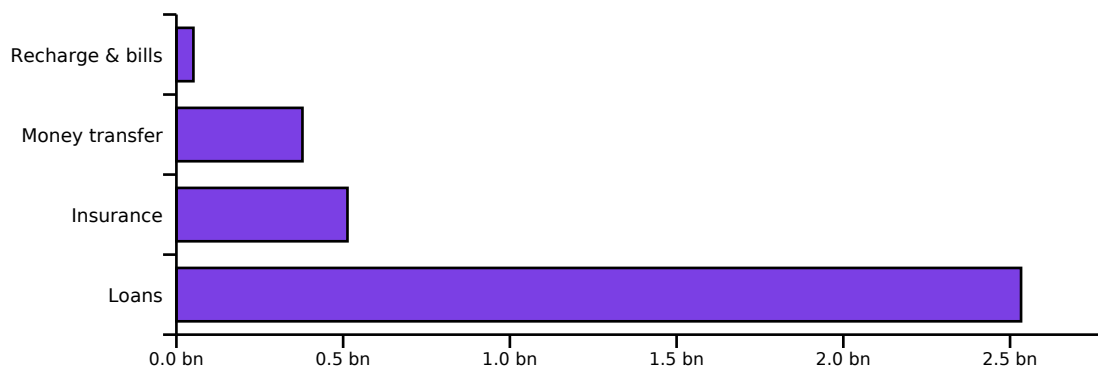
Validation note: The dashboard Top 5 Users visual groups by customer name, so common names combine multiple user IDs. For true customer-level ranking, use User ID as the unique key and show name only as a label.

03 | PERFORMANCE OVERVIEW

What the data revealed

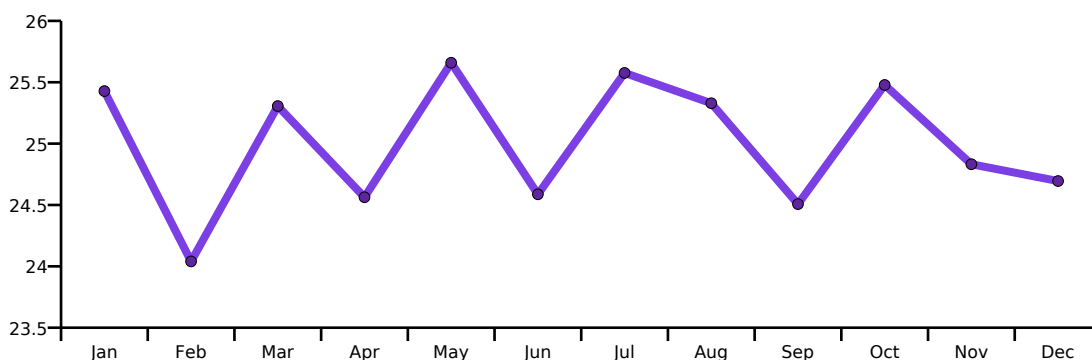
INR 11,581	287,993	12,007	4.0%
Average transaction	Successful payments	Unsuccessful payments	Unsuccessful rate

Transaction value by service



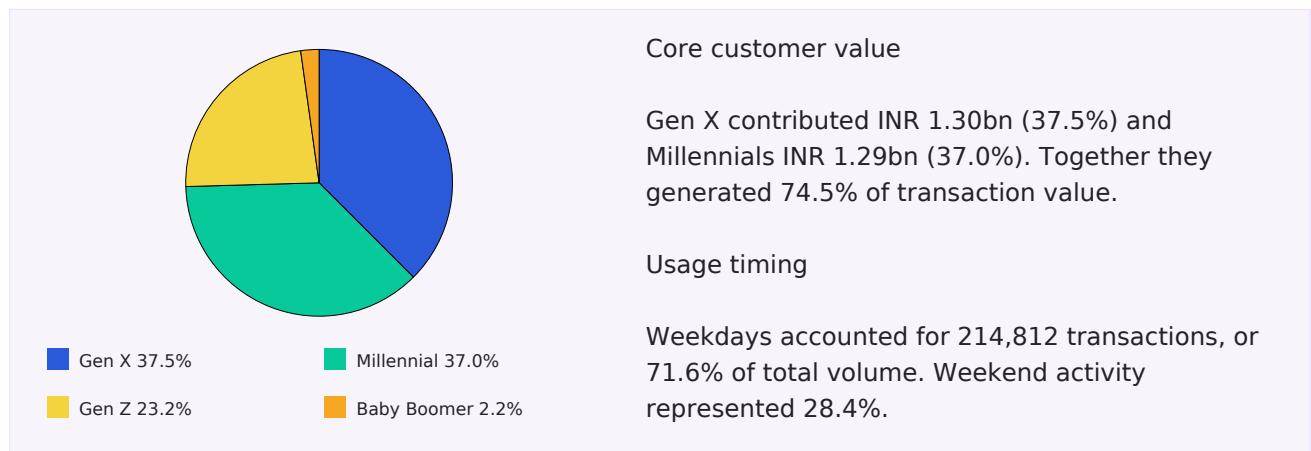
Loans dominate economic value. Loans generated INR 2.53bn, representing 72.9% of all transaction value from only 16.7% of transaction volume. Money transfer produced 50% of transactions but only 10.9% of value, showing a high-frequency, low-ticket use case.

Monthly transaction volume (thousands)



Demand is stable but not flat. Monthly volume remained between 24.0K and 25.7K. May had the most transactions (25,659), while July delivered the highest transaction value (INR 304.5m). February was the lowest month by both volume and value.

Who contributes value and where payments fail



Payment outcome and failure signals

Outcome / reason	Transactions	Interpretation
Successful	287,993	96.0% of all transactions
Server error reason	4,053	Largest unsuccessful reason by count
Wrong PIN reason	3,315	Customer authentication friction
Insufficient amount reason	3,304	Funding / balance constraint
Other unsuccessful reasons	1,335	Wrong Info and Bank Denied

Operational impact: Although the success rate is strong, unsuccessful transactions carry INR 141.3m in attempted value. Loan failures alone account for about INR 102.0m, so reliability work should be prioritised by value at risk, not only failure count.

From dashboard insight to business action

Question	Evidence from analysis	Business action
Where is most value generated?	Loans contribute INR 2.53bn, or 72.9% of total value.	Protect the loan journey, monitor risk and prioritise high-value service availability.
When is activity highest?	Weekdays represent 71.6% of transactions. May has the most transactions; July has the highest value.	Align operational capacity and campaigns with weekday and peak-month demand.
Which users drive value?	Gen X and Millennials contribute 74.5% of value combined.	Use segment-specific offers and retention journeys for the two core cohorts.
Where does the journey fail?	12,007 unsuccessful transactions. Server error is the largest failure reason by count.	Track failure reason, service and time to reduce avoidable drop-offs.

Recommended actions

Reduce technical payment failures

Create a failure-monitoring view by service, reason and time. Set alerts for server-error spikes and investigate high-value loan failures first.

Protect the loan journey

Track success rate, attempted value and failed value for loans separately. Prioritise uptime and recovery messaging because loans contribute 72.9% of value.

Use cohort-led targeting

Design retention and cross-sell campaigns for Gen X and Millennial users, while using lower-ticket adoption journeys for Gen Z.

Plan for demand peaks

Schedule support coverage and campaign activity around weekdays. Review May volume and July value drivers before repeating peak-period campaigns.

Improve customer identification

Use User ID, not name, for rankings and drill-through. This prevents different customers with the same name from being combined.

Deliverables and portfolio evidence

Deliverable	What it demonstrates
Clean analytical dataset	Validated IDs, dates, amounts and categorical fields across 407,658 source records.
Data model	User dimension connected to transaction fact data through User ID.
KPI measures	Transactions, transaction value, unique users, success rate and segment contribution.
Interactive Power BI dashboard	Month and payment-status filters, KPI cards, trends, service mix, age segments, top users and weekday/weekend analysis.
Business insight summary	Concise interpretation of value concentration, demand timing, customer contribution and payment reliability.
Recommendations	Prioritised actions for operations, product reliability, marketing and data governance.
Case-study PDF	Recruiter-friendly documentation of the business problem, process, results and impact.

Portfolio impact

This project shows the complete analytics cycle: converting raw customer and transaction data into a validated model, meaningful KPIs, interactive reporting and business recommendations.

Suggested portfolio title: PhonePe Transaction Analytics - Customer, Service and Payment Performance Dashboard

Suggested one-line description: Analysed 300K digital payment transactions worth INR 3.47bn to identify service value concentration, customer contribution, demand patterns and payment failure opportunities using Power BI.

Source: Phonepe-Final-Dataset(1).xlsx and the submitted Power BI dashboard screenshot. Figures are rounded for presentation.